Multimodal Fusion for Highlight Detection in Lifestyle Live Streams

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Abstract

*This research aims to address the challenge of automatically detecting and extracting highlights from hours-long lifestyle ("Just Chatting") live streams on Twitch. Traditional approaches primarily focus on gaming streams, where highlights often feature distinct audiovisual cues (*e.g. *kills, explosions) or rely solely on fluctuations in chat traffic. These methods struggle to capture the highlights in lifestyle streams, which are often shaped by complex conversations, situational humor, and audience interactions. To address this issue, this study proposes a novel model integrating three modalities: video, live stream transcripts, and chat content. Enhancing multimodal fusion is expected to improve the recognition of subtle humor and key interactions, thereby increasing the accuracy of highlight detection in lifestyle livestreams.*

1. Introduction

1.1. Motivation

Twitch has emerged as a top global entertainment platform, where "Just Chatting" streams, a category focused on unscripted, everyday interactions, now attract more viewing hours than game streams, demonstrating viewers' strong demand for authentic, high-quality interactive content. However, live streams often last for 4 to 8 hours, making it difficult for viewers to locate highlights on their own. While manual editing can address this issue, the process is time-consuming and labor-intensive. In addition, existing automated highlight detection tools are mostly designed for game streams and lack a deep understanding of audience interactions, limiting their effectiveness for conversation-driven lifestyle streams.

1.2. Problem Definition

This study aims to develop a model that automatically identifies highlight segments on full Twitch lifestyle VODs (including video, audio, transcripts, and timestamped chat), producing time intervals $[(start_1, end_1), ...]$ that closely match highlights selected by professional editors. The main contributions of this work are summarized as follows:

- A video-to-clip alignment method that synchronizes full-length VODs with corresponding highlight clips.
- A multimodal framework for live-stream highlight detection that fuses ASR, chat events, and visual features.
- A lightweight pre-filtering module that reduces computation by 15% while maintaining 98.4% highlight recall.

2. Related Work

Current highlight detection methods can be classified into three categories: audio-visual, audience reaction-based, and multimodal fusion-based. Preliminary multimodal models, such as UMT [5] and MH-DETR [8], comprehensively integrate video and audio but require explicit text-based queries and do not consider chat conversations, making them unsuitable for spontaneous livestreams. Audience reaction-based ones [2, 3] use chat frequency or sentiment, detecting "loud" moments but missing subtle highlights hidden within dialogue. Multimodal fusion-based models specialized in the domain of e-sports [1, 7] integrate video, audio, and chat features but are based on strong visual/audio cues, limiting their transferability to conversation-driven lifestyle streams. Recently, large language models (LLMs) have been explored for understanding long-form videos through textual reasoning. Chapter-LLaMA [6] demonstrates that LLMs can efficiently segment hour-long videos into coherent chapters by leveraging multimodal inputs. Similarly, Video Summarization with Large Language Models [4] integrates frame captions and transcripts to generate coherent summaries, showing that language-based reasoning can outperform traditional feature-based summarization.

3. Method

The model architecture of this study comprises three stages: feature extraction, cross-modal fusion and final highlight prediction. (as shown in Fig. 1)

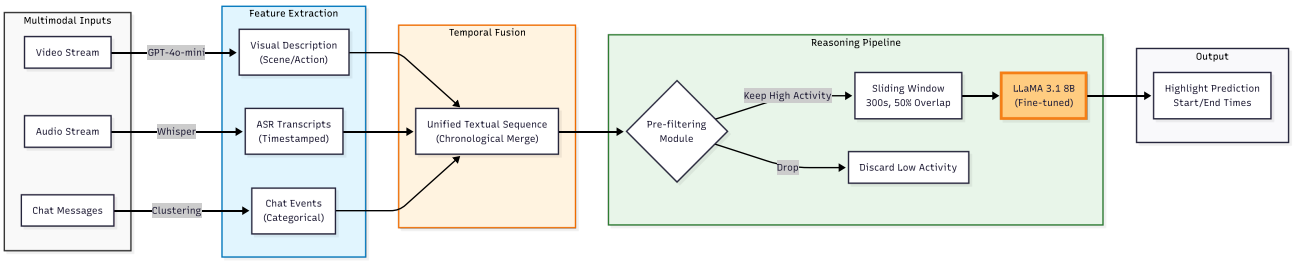

Figure 1. Overview of the dataset construction process. Twitch VODs and highlight clips are collected and aligned using perceptual hashing (pHash) and structural similarity (SSIM) to produce synchronized datasets for training.

Event Type	Description	Highlight Correlation
CHAT_SPIKE_LAUGH	Burst of laughter indicators (LOL, LMAO)	49%
CHAT_SPIKE_CONFUSED	Expressions of confusion (???, what, huh)	24%
CHAT_SPIKE_RELATIVE	References to streamer/content (she, he, they)	14%
CHAT_SPIKE_SPAM	Repeated messages or emotes	7.5%
CHAT_SPIKE_CLIP_MOMENT	Explicit "clip it" requests	1% (strong signal)
CHAT_CALM	Low activity periods	1.5% (negative signal)

Table 1. Categorical chat events and their correlation with highlight segments.

3.1. Dataset Construction

This study collects live streams from various non-gaming categories on Twitch, such as Just Chatting, Cooking, and Reaction, to construct a multimodal highlight detection dataset. Since highlight annotations are not readily available, they are generated semi-automatically through cross-platform content matching. Each highlight clip is aligned to its corresponding full-length VOD by segmenting the videos, extracting frame-level perceptual hashes (pHash), and computing Hamming distances with SSIM verification, as shown in Fig. 2. Video frames, speech transcripts, and chat messages are then preprocessed, temporally synchronized, and merged into a unified sequence to facilitate cross-modal reasoning and highlight prediction by the model.

3.2. Multimodal Event Representation

Feature representations from ASR, chat, and visual modalities are first extracted and then temporally aligned to produce a unified textual sequence (as shown in Fig. 3), which serves as the input for subsequent highlight reasoning.

ASR Processing

Automatic Speech Recognition (ASR) is performed using Whisper, generating timestamped transcript segments. Each ASR event is represented as:

- `timestamp_sec`: the time of the utterance
- `text`: the transcribed speech content

Chat Event Classification

Raw chat messages are noisy and voluminous. To mitigate noise while retaining semantically meaningful signals, chat messages are pre-processed into categorical events using keyword matching and temporal clustering. Table 1 summarizes the event types and their correlation with highlights.

Visual Description

Visual context is extracted using a two-step process. First, videos are segmented into semantic chapters based on ASR transcripts using GPT-4o-mini. Second, five representative frames per chapter are sampled and described using GPT-4o-mini's vision model. The visual descriptions encode scene context, streamer actions, and on-screen content, capturing the location, the streamer's behavior, and any visual elements displayed during the stream. Although visual events are fewer than ASR transcripts and chat events, they provide important information for identifying which scenes or actions are likely to elicit audience reactions and become highlight segments. The chapter-based frame sampling strategy ensures temporal coverage while maintaining computational efficiency.

3.3. Sliding Window Processing

Given the long duration of live streams, a sliding window approach is employed to segment videos into manageable units. Each window spans 300 seconds with a 50% overlap,

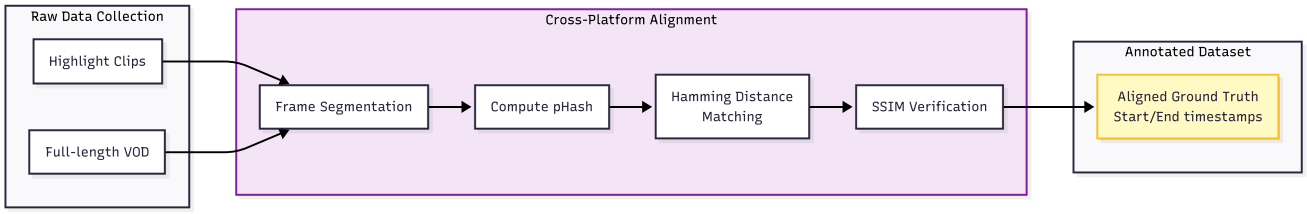

Figure 2. Overall architecture of the proposed multimodal highlight detection system. The pipeline integrates visual, speech, and chatroom modalities through preprocessing, temporal synchronization, and LLaMA-based highlight intensity prediction.

```
00:15:34 [ASR] Are you running the thing?
00:15:35 [VISUAL]
    1. Main Activity/Scene: A group of friends at a party near a waterpark or lake.
    2. People and Reactions: Foreground person is enthusiastic; others show relaxed.
    3. On-Screen Elements: Minimal graphics, including AT&T branding.
    4. Atmosphere and Energy: Vibrant, lively, and high-energy.
00:15:35 [CHAT_SPIKE_CONFUSED]
00:15:35 [ASR] Yes.
00:15:36 [ASR] Yeah, we were gonna run our burger shop.
```

Figure 3. Example of a merged text representation combining ASR, chat, and visual events. Events are aligned temporally to form a unified input for highlight detection.

capturing between 50 and 200 multimodal events depending on activity levels.

3.4. LLM-based Highlight Detection

The LLaMA 3.1 8B model is fine-tuned on the annotated multimodal dataset to perform highlight detection. The model parses the structured event timeline, identifies patterns indicative of highlight segments, and predicts precise temporal boundaries for each segment.

3.5. Pre-filtering Module

To reduce computational overhead, a lightweight pre-filtering module that scores video windows based on empirically derived signal weights is employed. The score for each window is computed as a weighted combination of chat, ASR, and visual features, along with event density. Windows corresponding to low-highlight activities, such as walking, shopping, or exercising, are penalized. If a window's score falls below the threshold, it will be discarded.

4. Experiment

The LLaMA 3.1 8B model is fine-tuned using QLoRA [12] on the annotated dataset to parse multimodal event timelines, identify patterns indicative of highlights, and output precise temporal boundaries.

4.1. Evaluation Metrics

Performance is assessed with standard temporal localization metrics, including Precision, Recall, and F1, computed under an IoU threshold of 0.3. These metrics quantify the proportion of predicted highlights that correctly match ground truth segments, the fraction of ground truth highlights detected, and their harmonic mean, respectively.

4.2. Main Results

The full model with pre-filtering (threshold set to 50) is evaluated on a validation set of four videos containing 91 annotated highlights across 487 windows. The system achieves a precision of 0.76%, recall of 12.33%, and F1 of 1.42%, detecting 28 true positives while generating 3,680 false positives. Pre-filtering reduces the number of windows processed from 487 to 411, corresponding to a 15.6% reduction in computational load, while maintaining 98.4% coverage of ground truth highlights. The model exhibits significant over-prediction, averaging 9.0 predicted highlights per five-minute window compared to 0.55 in the ground truth, resulting in reduced precision despite moderate recall. This issue is driven by excessive noise in ASR and other modalities, insufficient weighting of negative samples, and limited dataset coverage (single streamer, English-only). In addition, the latency introduced by sliding-window inference

Metric	Value	Count
Precision	0.76%	–
Recall	12.33%	–
F1 Score	1.42%	–
True Positives	–	28
False Positives	–	3,680
False Negatives	–	199
Avg. Inference Time	37.1s/window	–
Total Evaluation Time	253.8 min	411 windows

Table 2. Overall performance of our full model with pre-filtering (threshold=50) on the validation set (4 videos, 91 ground truth highlights, 487 windows).

Configuration	Precision	Recall	F1
ASR Only	1.0%	17.2%	1.8%
Chat Only	3.8%	28.3%	6.7%
Visual Only	4.6%	8.1%	5.9%
ASR + Chat	1.2%	17.2%	2.2%
ASR + Visual	1.4%	22.2%	2.6%
Chat + Visual	2.8%	22.2%	4.9%
Full (All Modalities)	1.1%	15.2%	2.1%
Full + Pre-filter	1.4%	15.9%	2.6%

Table 3. Ablation study results on the validation set, evaluating the contribution of each modality and the pre-filtering module.

further constrains generalization. Future work should focus on rebalancing negative samples, expanding the dataset through large-scale annotation, and incorporating feature-level denoising techniques to reduce modality noise and improve highlight detection accuracy.

4.3. Ablation Study

To examine the contribution of individual modalities, a series of ablation experiments is conducted. Single-modality models based on ASR, chat, or visual descriptions demonstrate limited predictive power, as summarized in Table 3.

The ablation study highlights the contribution of each modality to highlight detection. Surprisingly, the model using only chat messages achieves the highest F1 score (6.7%), with a precision of 3.8% and recall of 28.3%, outperforming models that rely solely on ASR or visual features. This is likely due to the high-quality preprocessing of chat events (e.g., LAUGH, CONFUSED), which provides sparse but reliable signals. In contrast, the ASR-only model exhibits severe over-prediction (F1=1.8%), generating many false positives due to noisy audio signals, indicating limited discriminative power. Interestingly, the full multimodal model achieves a lower F1 score (2.1%) than the chat-only model, reflecting the challenges of over-prediction and the difficulty of correctly integrating noisy ASR features with the clean chat signals. These results confirm that the current training approach fails to effectively learn multimodal fusion, highlighting the need for improved strategies to fully leverage complementary signals.

5. Conclusion

Multimodal fusion under the current single-stage QLoRA fine-tuning framework fails to properly weight modality contributions, causing ASR noise to overwhelm clean chat signals and limiting the effectiveness of the full model. Pre-filtering offers a modest improvement by reducing low-quality windows, maintaining high recall while slightly increasing precision. Visual features contribute complementary context but have limited predictive power alone and can slightly degrade chat-only performance when combined. Future work should explore modality-aware attention, negative sample mining, and progressive fusion strategies to fully leverage high-quality signals while mitigating noise from less reliable modalities.

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